

From Analytics to Action

Interactive Dashboard & Capstone Presentation

Maven Fuzzy Factory order analytics | 32,313 orders | 2012-03-19 to 2015-03-19

DECISION LENS

TOTAL REVENUE

Current analytical base
\$2,626,811.51

SCALING

Volume growth +
product mix

PROTECTING

Margin quality +
data governance

Executive snapshot

Five headline measures from the cleaned order dataset, followed by the implications for decision makers.

TOTAL REVENUE

\$2,626,811.51

Across full dataset

TOTAL ORDERS

32,313

Unique order records

AVERAGE ORDER VALUE

\$81.29

Revenue / order

TOTAL PROFIT

\$1,657,781.50

Revenue - total cost

PROFIT MARGIN

63.11%

Profit / revenue

What the data says

- Revenue and order volume expand sharply through 2014, the strongest full year in the dataset.
- Profit margin improves from 61.0% in 2012 to 63.5% in 2014 and 63.7% in the 2015 partial period.
- Product 1 accounts for 73.6% of revenue, creating a clear concentration dependency.
- 76.1% of orders contain one item, leaving an evidence-based AOV / basket-size opportunity.

Decision focus

- 1 Raise basket value without eroding margin quality.
- 2 Reduce dependence on a single product driver.
- 3 Use KPI governance so future decisions are repeatable.

Business question & analytical scope

The capstone translates the work completed in Tasks 03-05 into an executive decision narrative.

01

How is revenue scaling?

Track revenue, orders and AOV across time.

02

Where is profit created?

Compare profit and margin with price and COGS.

03

What drives performance?

Use correlations and product contribution to identify drivers.

04

What should management do?

Turn the evidence into focused actions and a 90-day plan.

Evidence base

32,313 rows | 15 fields | 2012-03 to 2015-03 | cleaned order-level dataset

Core fields: orders, user/session IDs, primary product ID, items, price, COGS, derived revenue, total cost, profit and margin.

Scope limitation

The available dataset does not contain CAC, churn or geographic fields. Those measures are therefore not fabricated or used as evidence in this report.

Methodology: from raw data to decisions

A traceable workflow connects the cleaned dataset to statistical evidence and the interactive dashboard.



Quality controls retained from Task 03

No missing values | No duplicate rows | Valid positive quantity / price / COGS | Valid date range | IQR outliers reviewed as business observations

Revenue and order growth

2014 is the strongest full year in the dataset; 2015 contains only Jan-Mar records and should not be compared as a full-year total.

2014 revenue

\$1,583,913.77

2014 orders

16,860

2015 partial period

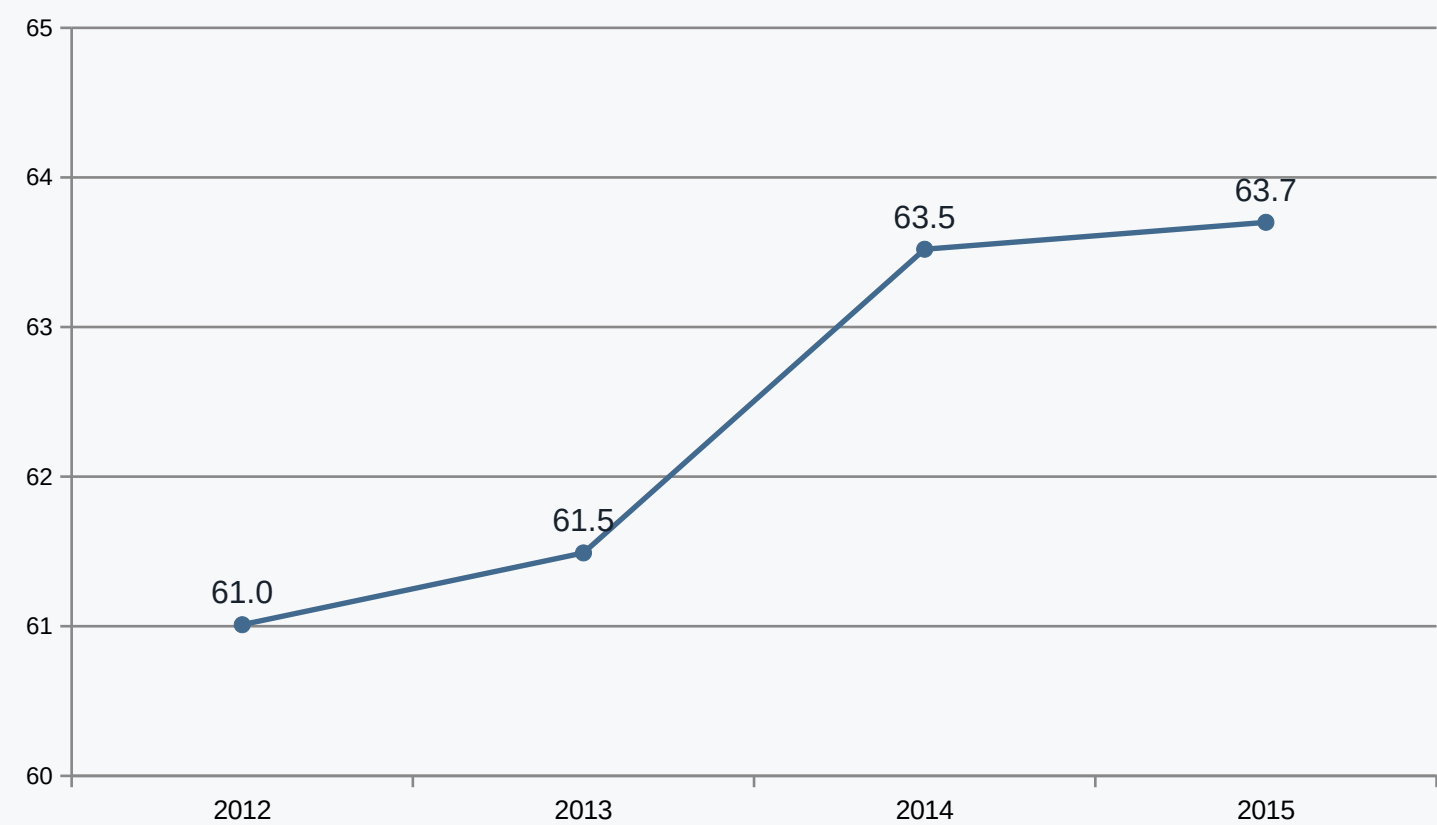
Jan-Mar only

Interpretation

Scale-up through 2014 is the clearest historical signal. Use period-normalized metrics for any future benchmarking.

Profitability and margin quality

Profit expands alongside revenue while margin remains comparatively stable, moving upward over the observed period.



TOTAL PROFIT

\$1,657,781.50

Across full dataset

CURRENT-BASE MARGIN

63.11%

Full-dataset weighted margin

TOTAL COST

\$969,030.01

Revenue less profit

Implication: protect margin as scale increases.

Statistical evidence: three tested relationships

Pearson correlation tests from Task 04 identify strong linear associations; they do not establish causation.

Price vs Revenue

$r = 0.9783$

$p < 0.001$

Very strong positive
association

Price vs Profit

$r = 0.9731$

$p < 0.001$

Very strong positive
association

COGS vs Profit

$r = 0.9270$

$p < 0.001$

Strong positive
association

How to use this evidence

The relationships support three practical decisions:

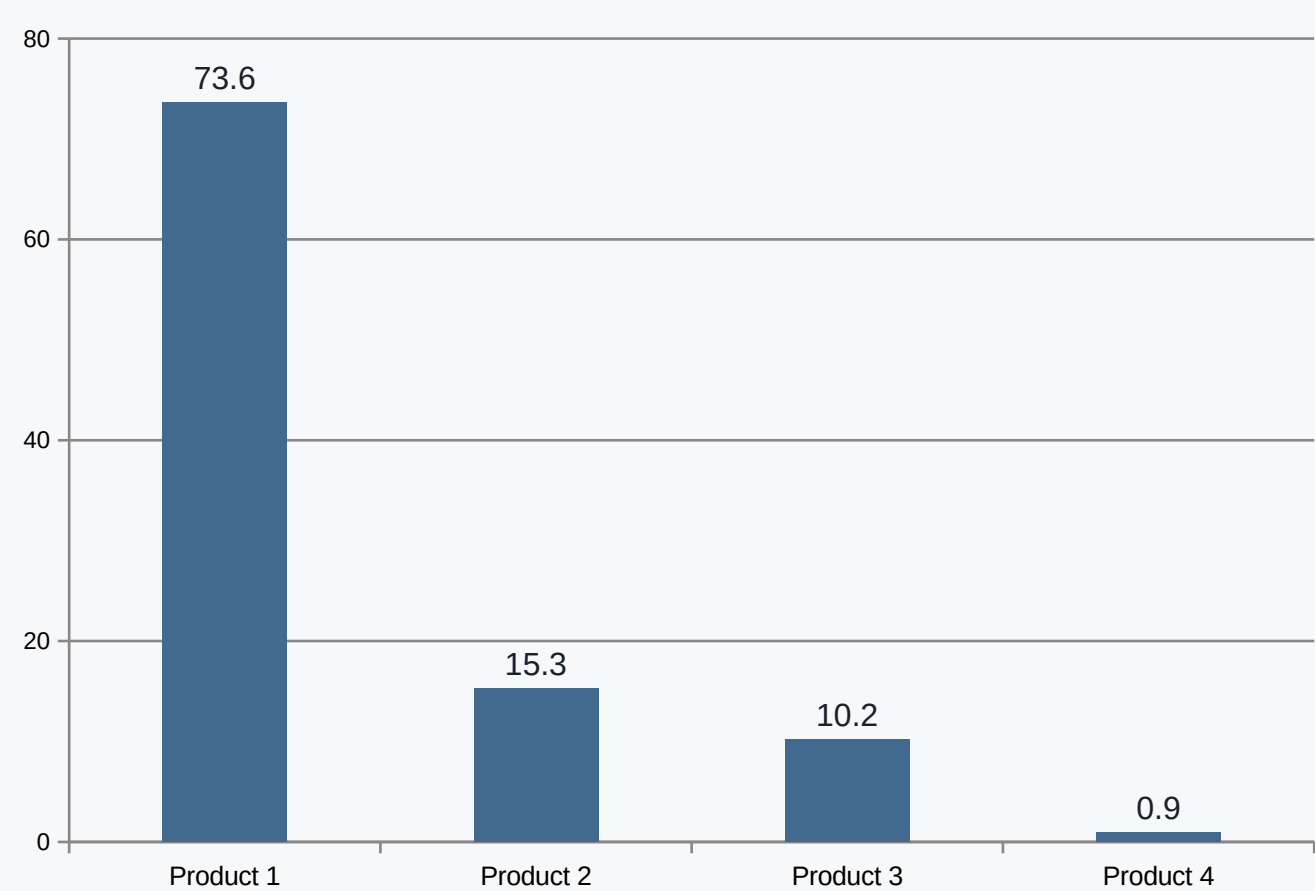
- 1 Use pricing and basket strategy to grow revenue.
- 2 Manage COGS carefully as volume scales.
- 3 Validate every intervention with controlled measurement.

Statistical note

All three tests return p-values below 0.05. The findings indicate association in this dataset, not a causal effect.

Revenue drivers: mix matters

Product contribution and basket composition point to concentration risk alongside an AOV opportunity.



Product concentration

73.6% of revenue comes from Product 1.

Basket composition

76.1% of orders contain one item.

Opportunity: test bundles, cross-sell and threshold-based offers while tracking margin.

Risk areas and management watchpoints

The same signals that support growth also define where management should monitor downside exposure.

HIGH	Product concentration	73.6% of revenue is tied to Product 1.	Watch: Diversify contribution by validating adjacent products and bundles.
HIGH	Basket-size dependence	76.1% of orders contain one item.	Watch: Test AOV levers before assuming traffic growth is the only path to scale.
MEDIUM	Period comparability	2015 is a partial Jan-Mar period.	Watch: Use like-for-like time windows for performance benchmarking.
MEDIUM	Measurement scope	CAC, churn and geography are absent from the source data.	Watch: Add governed fields before these KPIs drive executive decisions.



Three strategic recommendations

Recommendations are directly tied to observed revenue, profitability, product-mix and basket signals.

01

Increase basket value

Pilot bundles, cross-sell prompts and order-value thresholds.

Primary KPI: AOV | Guardrail: profit margin

LOW-MEDIUM

Implementation effort

02

Broaden product contribution

Create targeted merchandising / bundle tests around Products 2-4 to reduce dependency on Product 1.

Primary KPI: revenue share by product | Guardrail: conversion

MEDIUM

Implementation effort

03

Protect margin through scale

Set a monthly COGS and margin review with controlled pricing / cost tests.

Primary KPI: profit margin | Guardrail: revenue growth

LOW

Implementation effort

Illustrative ROI scenario + 90-day implementation

ROI is a scenario, not a forecast. It uses explicit assumptions against the current dataset revenue base.

Scenario economics

Assumption 5% revenue uplift + 1.0 percentage-point margin improvement

Incremental contribution \$109,157.19

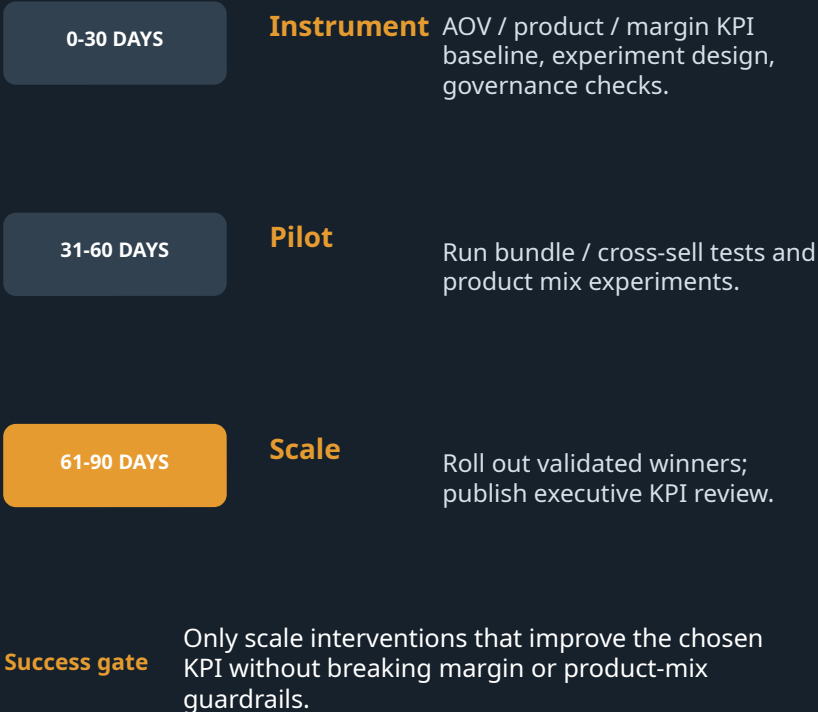
Illustrative implementation cost \$30,000.00

Illustrative ROI **264%**

Calculation: $ROI = (incremental\ contribution - implementation\ cost) / implementation\ cost$.

Use pilot results to replace assumptions with observed economics.

90-day implementation timeline



Final portfolio proof

The capstone is supported by the completed analytical work and the deployed interactive dashboard.

PUBLIC PROOF LINKS

Task 03 - Data Cleaning

<https://github.com/Gautamdutta-star/task-03-data-cleaning-preprocessing>

Task 04 - EDA

<https://github.com/Gautamdutta-star/task-04-exploratory-data-analysis>

Task 05 - Dashboard

<https://github.com/Gautamdutta-star/task-05-interactive-dashboard>

Live Streamlit Dashboard

<https://task-05-interactive-dashboard-kggxg5b4p8kweaovaohvmm.streamlit.app/>

Source chain: cleaned CSV -> EDA notebook -> interactive dashboard -> executive decision report

Executive decision summary

- Scale the levers that increase basket value while protecting margin.
- Diversify product contribution to reduce dependence on Product 1.
- Formalize KPI governance and experimentation before broad rollout.

Capstone takeaway

Move from descriptive reporting to measurable, governed business experiments.

End of report